A Resource Allocation Model for Childhood Vaccination in Michigan

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Problem Statement

Full vaccine series in children aged 19-35 months are crucial to community health. Across the state of Michigan, only 97,617 out of 146,682 children 19-35 months have completed the recommended 4313314 vaccine series, which is made up of DTaP, Polio, MMR, Hib, HepB, Varicella, and PCV. This 66.55% coverage rate falls far short of the recommended 90-95% coverage to achieve herd immunity for the most contagious diseases in the series. Low coverage is driven by barriers including lack of publicity, education, availability, and accessibility of public health services. In this project, a mathematical integer programming model is designed and solved to minimize the number of children 19-35 months old who have not completed the 4313314 vaccine series across the state of Michigan by allocating a budget of mobile vaccination clinics.

Data Needed to Solve the Problem

This model requires vaccination coverage data and population data for each of Michigan's 83 counties and Detroit. For vaccination coverage, the model uses the 4313314 series completion rate for children aged 19–35 months, which captures whether each child has received all vaccines in the recommended childhood series. This age range is specifically targeted because it represents the window in which children should have completed the full series according to the CDC immunization schedule. For population data, the model requires the total number of children aged 19–35 months in each county, which is necessary to convert coverage rates into absolute counts of unvaccinated children. County latitude and longitude coordinates are also used for geographic visualization of the allocation results in Power BI.

Data Source

The vaccination coverage data is sourced from the MDHHS County Immunization Report Cards, a publicly available dashboard maintained by the Michigan Care Improvement Registry (MCIR). This dataset provides quarterly county-level coverage estimates for all recommended childhood and adolescent vaccine series and was accessed for Q4 2025. The 19–35 month old population denominators are not included in the public dashboard. That data was obtained directly from MDHHS by request, as the registry maintains enrollment records for age-eligible children by county. The county coordinate data was sourced from publicly available Michigan GIS records.

Mathematical Model

This model is formulated as an integer program to optimally allocate a fixed budget of 3000 mobile vaccination clinic visits across Michigan’s 83 counties and Detroit. The objective

function minimizes the total weighted unvaccinated children after deployment, where each county's residual unvaccinated population is weighted by its incomplete rate. This prioritizes counties with the lowest existing coverage. The per-visit vaccination yield is county-specific and scaled by existing coverage. This reflects the assumption that residents of lower-coverage counties are harder to reach, whether because of geographic barriers or vaccine hesitancy, and therefore fewer children will be vaccinated in those counties. Each county must receive enough aid to improve their vaccine coverage by at least 5% to ensure that no county will be ignored by the solver. Furthermore, each county is capped at receiving enough visits to bring its population to 90% covered. This reflects the real-world reality that the last few percent of vaccine coverage are the hardest to achieve and often deploying resources to that final margin comes at an exponentially diminishing return.

Note: This model was solved using Python and Google OR-Tools CP-SAT solver. See Appendix, Link 1 for a link to the Github repository for this model.

Model:

$$MINIMIZE \quad Z = \sum_{i=1}^{84} w_i(U_i - \alpha_i x_i)$$

$$SUBJECT\ TO \quad \sum_{i=1}^{84} x_i \leq B$$

$$x_i \geq \left\lceil \frac{0.05 \cdot U_i}{\alpha_i} \right\rceil \ \forall i \quad \text{(5\% equity floor)}$$

$$x_i \leq \left\lfloor \frac{0.90 \cdot U_i}{\alpha_i} \right\rfloor \ \forall i \quad \text{(90\% herd immunity cap)}$$

$$x_i \geq 0 \ \forall i$$

Decision Variable:

- x_i = clinic visits allocated to county i

Parameters:

- $w_i = 100 - covg_i$ = incomplete rate weight
- $covg_i$ = existing coverage rate in county i
- P_i = population aged 19-35 months in county i
- $U_i = P_i \cdot \frac{100 - covg_i}{100}$ = unvaccinated children in county i
- $\alpha_{max} = 20$ = max vaccinations per visit
- $\alpha_i = round(\alpha_{max} \cdot \frac{covg_i}{100})$ = children vaccinated per visit, weighted by current coverage rates
- B = 3000 = total visit budget

Results

Note: For the full output table of the solver for B = 3000, see Appendix, Table 2.

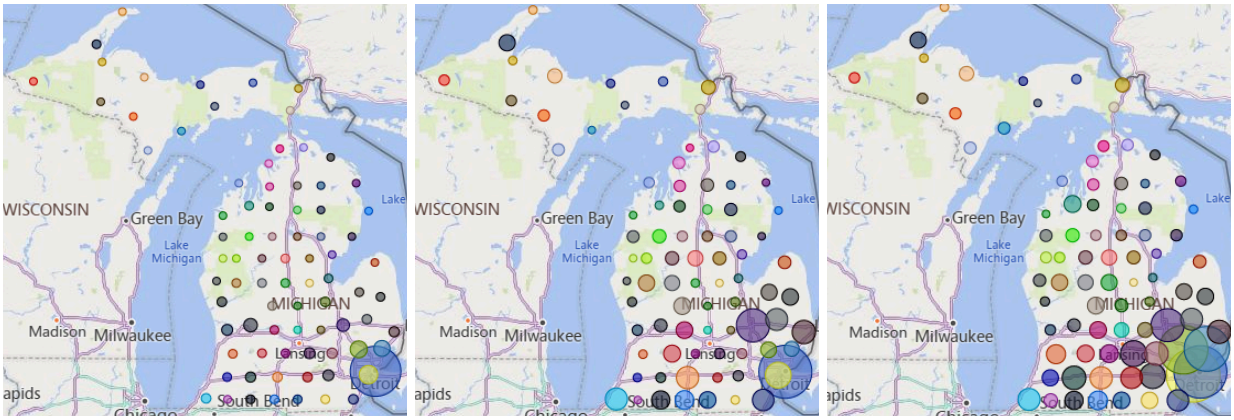
The solver minimizes the weighted sum of unvaccinated children remaining. This is each county's remaining unvaccinated population multiplied by its priority weight. Because lower-coverage counties carry higher weights, this value reflects not just how many children remain unvaccinated, but how inequitably that burden is distributed. Since this doesn't directly represent actual numbers, it is important to analyze the unweighted number of unvaccinated children as well. The weighted score is inflated above the raw headcount because the hardest-to-reach counties contribute disproportionately to the objective.

With a budget of 3,000, Z = 328,439. After deployment, there are only 10,853 unvaccinated children remaining, a stark difference from the original 49,065 unvaccinated. The model allocated 3,000 visits to vaccinate 38,212 children, a 77.9% reduction in the unvaccinated backlog. Combined with the 97,617 children already vaccinated prior to deployment, this brings the total vaccinated population to 135,829, representing 92.6% of all 19-35 month olds in Michigan, which surpasses the 90% herd immunity threshold.

From a sensitivity analysis standpoint, it is interesting to view how the model handles other budgets. With a budget of 500, Z = 1,452,127. The 500-visit allocation vaccinates 5,936 children (12.1% reduction), bringing total statewide coverage to 70.6%, very far short of herd immunity. With a budget of 1,500, Z = 972,993. The 1,500-visit allocation vaccinates 17,915 children (36.5% reduction), bringing total coverage to 78.7%, still below the threshold. Only at B = 3,000 does Michigan cross the 90% herd immunity threshold.

Budget	Z	Vaccinated	% Reduction of Unvaccinated	Remaining	% of Children Vaccinated	Herd Immunity?
500	1,452,127	5,936	12.1%	43,129	70.6%	NO
1,500	972,993	17,915	36.5%	31,150	78.7%	NO
3,000	328,439	38,212	77.9%	10,853	92.6%	YES

The following images display the results of the solver in a visual format. Each differently colored bubble on the map represents a county, and the size of the bubble represents the number of visits allocated to that county.



B = 500 *B = 1500* *B = 3000*

One interesting observation about the result is that Ontonagon County received 0 visits, even though the model requires each county to address 5% of their population. This occurs because the equity floor is overridden by the 90% herd immunity cap. Ontonagon has only 10 unvaccinated children, but its county-specific visit effectiveness $\alpha_i = 15$, meaning a single visit would vaccinate 15 children. Allocating a single visit to Ontonagon would cause its vaccinated population to exceed 90%. Since the cap sets $x_i \leq \lfloor \frac{0.90 \cdot U_i}{\alpha_i} \rfloor = \lfloor \frac{0.90 \cdot 10}{15} \rfloor$ maximum visits, the constraint becomes $x_i \leq 0$, effectively exempting the county from the equity floor. This is a limitation of the model, as counties with very small unvaccinated populations relative to their α_i size cannot be reached under the current formulation of the model.

Further, counties with above-average coverage rates, such as Kent, Ottawa, Saginaw, and Muskegon, receive only the minimum allocation and their coverage rates only increase between one or two percent, despite being among the more populous counties in the dataset. This is correct model behavior, as the above-average original coverage rates correlate with a lower w_i and higher α_i. Kent's w_i of 24.5 is less than half that of a county like Clare at 47.5, meaning the solver correctly treats each unvaccinated child in Clare as nearly twice as urgent as one in Kent.

Discussion of Results

The results above show that the model performs as intended, but a deeper analysis of the solution reveals some important insights. The shadow price on the budget constraint is 392, meaning that increasing the mobile clinic visit budget by one unit from 3,000 to 3,001 reduces the weighted unvaccinated objective by 392. Reduced costs are not applicable in this solution because when the budget is 3,000 the solver sends at least one visit to every county, meaning every decision variable receives at least its minimum equity floor allocation. Ontonagon having zero visits is forced by the constraints so it isn't valid to analyze as a reduced cost, because

unless the formula for α_i changes, Ontonagon will continue to be allocated zero visits, as discussed in the Results section.

There are some limitations of this model. First, α_i is calculated based on an assumption that resources allocated are less effective in lower-coverage counties. There isn't a perfect way to calculate the effectiveness of resources. This model also doesn't take into account travel costs. On one hand, ignoring travel costs is a limitation because it treats a visit to Keweenaw County the same as a visit to Ingham County, while in practice the budget would be partially consumed by travel because they are so far from each other. On the other hand, including travel costs would bias the model towards densely populated counties near the deployment hub and away from rural high-need counties, which would undermine the equity goal of the model. Furthermore, the budget constraint utilized in this model isn't derived from any real MDHHS operational figure. The budgets demonstrated in this report were chosen to demonstrate the model's ability to allocate resources across a variety of situations, but in reality, deployment would require real budget parameters. Finally, the vaccination coverage data given by the MDHHS is not granular enough to understand how many children may still not have received the full vaccine series even after deployment. For example, children who receive 1 out of 4 doses of DTaP from the mobile vaccination clinics are counted as 'vaccinated' by the model even though they have not actually completed the full series. The model treats any child reached by a visit as successfully vaccinated, when in reality completing the full multi-dose series often requires multiple appointments over time.

These results suggest that the MDHHS would achieve the greatest reduction in vaccination inequity by concentrating mobile clinic resources in high-population, low-coverage areas. Wayne, Oakland, Macomb, and Detroit collectively receive 1,495 visits, almost 50% of the budget, reflecting their large unvaccinated population. From a policy standpoint, this model supports a need-weighted deployment strategy rather than a population-weighted one. The sensitivity analysis further reveals strong returns on investment in the 500 to 3,000 visit range. Increasing the budget of 500 to 3,000 causes the share of unvaccinated children who are reached to jump from 12.1% to 77.9%. Future iterations of this project could include travel costs and multi-period planning, along with validated budgets and α_i values, and a more granular model to track dose-level progress.

Appendix

Table 1:

Herd Immunity Rates for 4313314 Vaccination Series

Vaccine	DTaP	Polio	MMR	Hib	HepB	Varicella	PCV
CDC Herd Immunity Coverage	92-94%	80%	93-95%	N/A	N/A	85-90%	N/A

Link 1:

https://github.com/blujaee/mi_vaccination_project

Table 2:

Complete solver output with B = 3000

County Name	Visits Allocated	α (estimated vaccinations per visit)	Previous Coverage (%)	New Coverage (%)
Alcona	2	14	70.8	95.58
Alger	2	12	59.57	85.11
Allegan	35	14	70.86	96.96
Alpena	5	15	73.2	94.81
Antrim	7	13	63.83	96.1
Arenac	3	13	64.6	88.82
Baraga	2	13	64.55	88.18
Barry	25	12	62.27	95.27
Bay	2	15	76.3	78.66
Benzie	1	14	72.25	79.58
Berrien	52	13	62.86	96.03
Branch	33	11	52.53	94.35
Calhoun	53	13	62.57	96.15
Cass	12	13	65.96	95.96
Charlevoix	8	12	61.46	93.36
Cheboygan	7	12	59.77	91.95
Chippewa	14	11	56.14	93.25
Clare	18	10	52.47	94.82
Clinton	16	15	73.48	96.71
Crawford	3	14	67.86	92.86

Delta	8	14	69.48	95.77
Detroit	416	11	52.99	95.24
Dickinson	7	13	65.26	92.75
Eaton	29	14	69.39	96.12
Emmet	6	14	70.22	93.82
Genesee	143	13	65.34	96.44
Gladwin	11	11	55.89	92.45
Gogebic	5	11	55.33	92
Grand Traverse	25	14	68.83	96.59
Gratiot	10	14	71.25	95.94
Hillsdale	24	11	56.2	94.74
Houghton	22	11	53.74	94.9
Huron	10	13	63.39	95.33
Ingham	80	14	68.59	96.76
Ionia	24	13	65.2	95.7
Iosco	7	13	65.83	94.36
Iron	5	10	50.41	91.74
Isabella	21	13	65.58	96.29
Jackson	49	13	67.16	96.46
Kalamazoo	62	15	72.94	97.21
Kalkaska	7	12	62.22	93.33
Kent	10	15	75.5	76.79
Keweenaw	2	7	33.33	75.76
Lake	5	11	55.07	94.93
Lapeer	51	10	50.35	94.93
Leelanau	5	13	62.9	92.31
Lenawee	23	14	68.74	95.87
Livingston	61	13	66.5	96.55
Luce	3	10	50	94.12
Mackinac	4	11	56.14	94.74
Macomb	295	13	65.8	96.48
Manistee	8	12	58.71	95.08
Marquette	15	14	68.46	96.06
Mason	6	14	69.94	95.71
Mecosta	18	12	60.52	95.47
Menominee	8	12	61.02	93.56

Midland	1	15	75.25	76.53
Missaukee	5	13	63.76	92.14
Monroe	48	13	62.73	96.1
Montcalm	25	13	63.85	96.13
Montmorency	2	12	62.37	88.17
Muskegon	3	14	71.61	73.18
Newaygo	22	12	60.08	95.09
Oakland	352	14	69.56	96.93
Oceana	9	13	66.17	95.27
Ogemaw	7	13	63.76	95.47
Ontonagon	0	15	77.27	77.27
Osceola	12	12	57.96	93.78
Oscoda	9	8	37.98	93.8
Otsego	10	12	59.82	95.54
Ottawa	5	14	72.06	73.56
Presque Isle	4	12	62.16	94.59
Roscommon	5	13	65.24	93.13
Saginaw	3	14	72.22	73.74
Sanilac	23	10	49.9	93.88
Schoolcraft	1	15	74.19	90.32
Shiawassee	17	14	70.37	96.01
St. Clair	60	12	61.94	95.65
St. Joseph	31	11	57	94.89
Tuscola	21	12	59.46	95.06
Van Buren	22	14	69.3	96.32
Washtenaw	74	14	71.72	93.77
Wayne	432	13	66.14	96.61
Wexford	12	13	62.55	94.65